HW - Voronoi Diagram Generator

35 Points Possible

10/14/2025

Attempt 1

10/14/2025

NEXT UP: Review Feedback

Attempt 1 Score: N/A

Add Comment

Unlimited Attempts Allowed

Details

Description

Create a C++ GUI application that displays a Voronoi diagram on an 800×600 canvas.
The program should include two main components:

- A button for loading a configuration file (`config.txt`)
- A canvas for displaying the generated diagram

When the button is pressed, a file dialog box should appear that allows the user to choose the config file.
Each line in the configuration file should contain a pair of integer values representing a coordinate point within the canvas.

After loading the file, the GUI should display the corresponding Voronoi diagram, where the number of regions is determined by the number of coordinate points specified in the input file.

Submission

Submit a Google Drive link (restrict access to DLSU accounts) containing the following

- Source code
- Screen recording showing the building and running of your code on Visual Studio / VSCode
  You must demonstrate 2 test cases
  Test case 1

```
242 410
372 158
57 130
150 479
362 55
```

Test case 2

```
469 194
430 284
358 195
579 143
636 447
94 179
238 45
632 213
723 475
672 389
```

Hints

- To generate a Voronoi diagram:
  Check every pixel/point in the diagram and compare it's Euclidean distance between all the "seed" points. The closest seed point determines the region or color of a pixel/point. You can randomly set the color assigned to the seed points.
- You can use QFileDialog for loading the config file

```
QString fileName = QFileDialog::getOpenFileName
(
    &window,
    "Load configuration file",
    "",
    "Text Files (*.txt)"
);

if (fileName.isEmpty())
    return;

std::ifstream file(fileName.toStdString());

// parse the file
```

- You may use QWidget or QImage for drawing the diagram

```
QImage image(width, height, QImage::Format_RGB3
2);

...
image.setPixelColor(x, y, color);
...
```

Grading

GUI	15 pts Complete	10 Some components are missing or not properly implemented	5 Most components not working or incorrectly implemented	0 No meaningful implementation

Correctness	20 All required features work correctly	15 Most features work correctly but with minor issues	8 Significant issues with program behavior	0 No meaningful implementation

https://drive.google.com/drive/folders/1iuXRfxC2CHWZ0HawAfhWocULvWQNaf_I?usp=sharing (https://drive.google.com/drive/folders/1iuXRfxC2CHWZ0HawAfhWocULvWQNaf_I?usp=sharing)

New Attempt